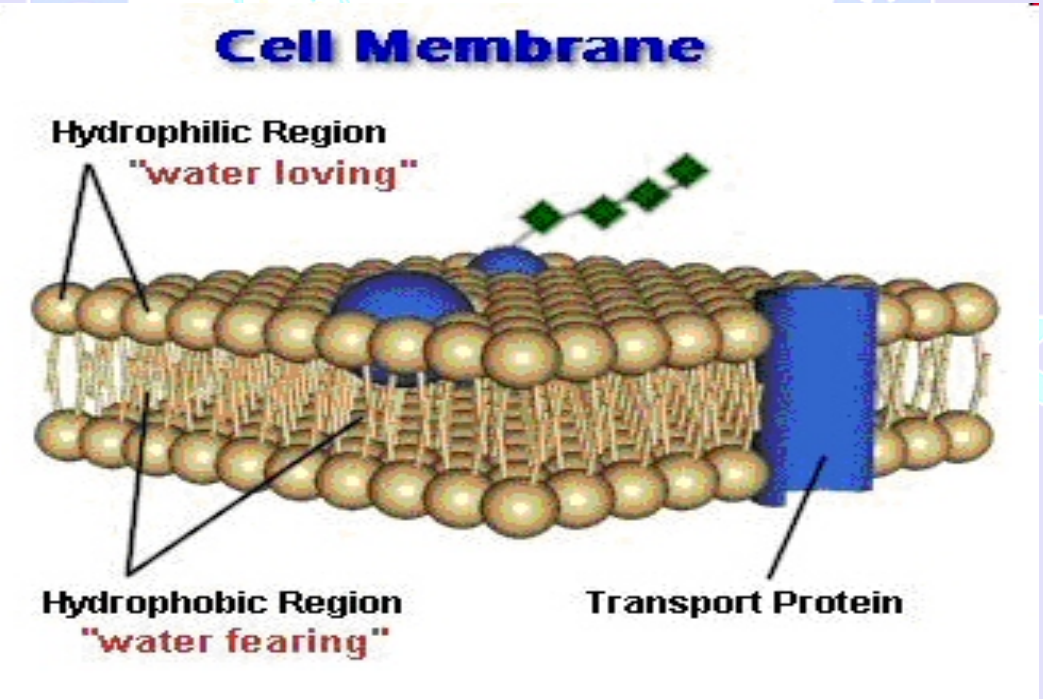
CELL MEMBRANE



LEARNING OBJECTIVES

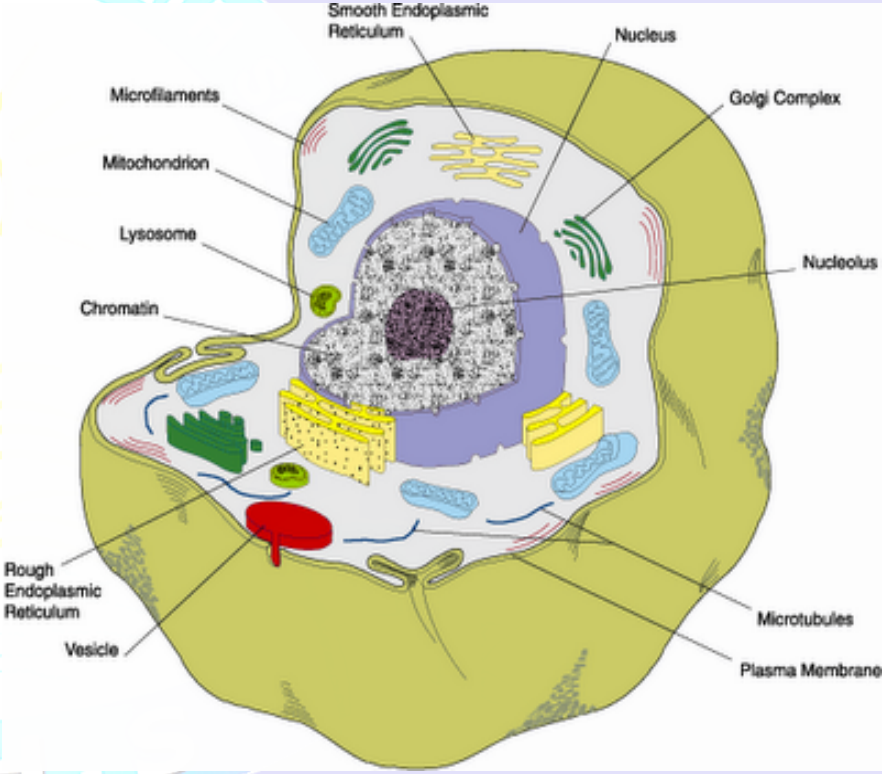
At the end of lecture the student of first semester should be able to:

- Identify the structure of Cell membrane
- Recognize the Phospholipid bilayer and its composition
- Differentiate between different types of Cell membrane proteins and carbohydrates and its functions.
- Identify electron microscopic features of cell membrane.
- Express fluid Mosaic Model of cell membrane.

CELL

- Living substance of plants and animals is known as <u>protoplasm</u>.
- The smallest unit of the protoplasm capable of independent existence is known as a <u>cell</u>.
- <u>Components of cell</u>.

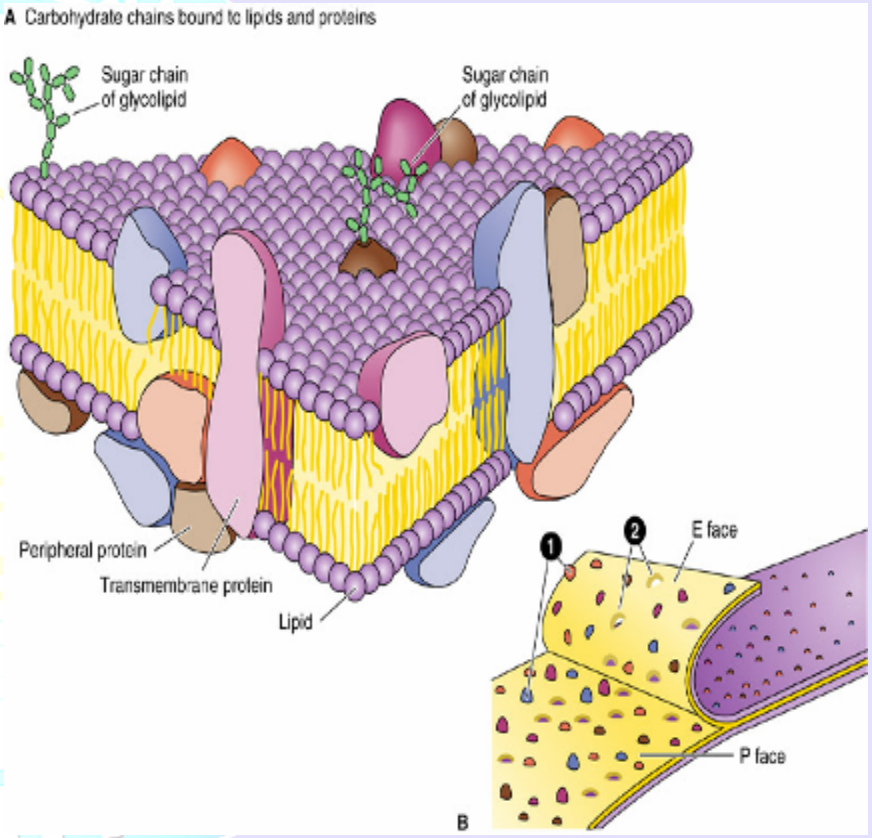
1. Cell membrane.
2. Nucleus.
3. Cytoplasm.



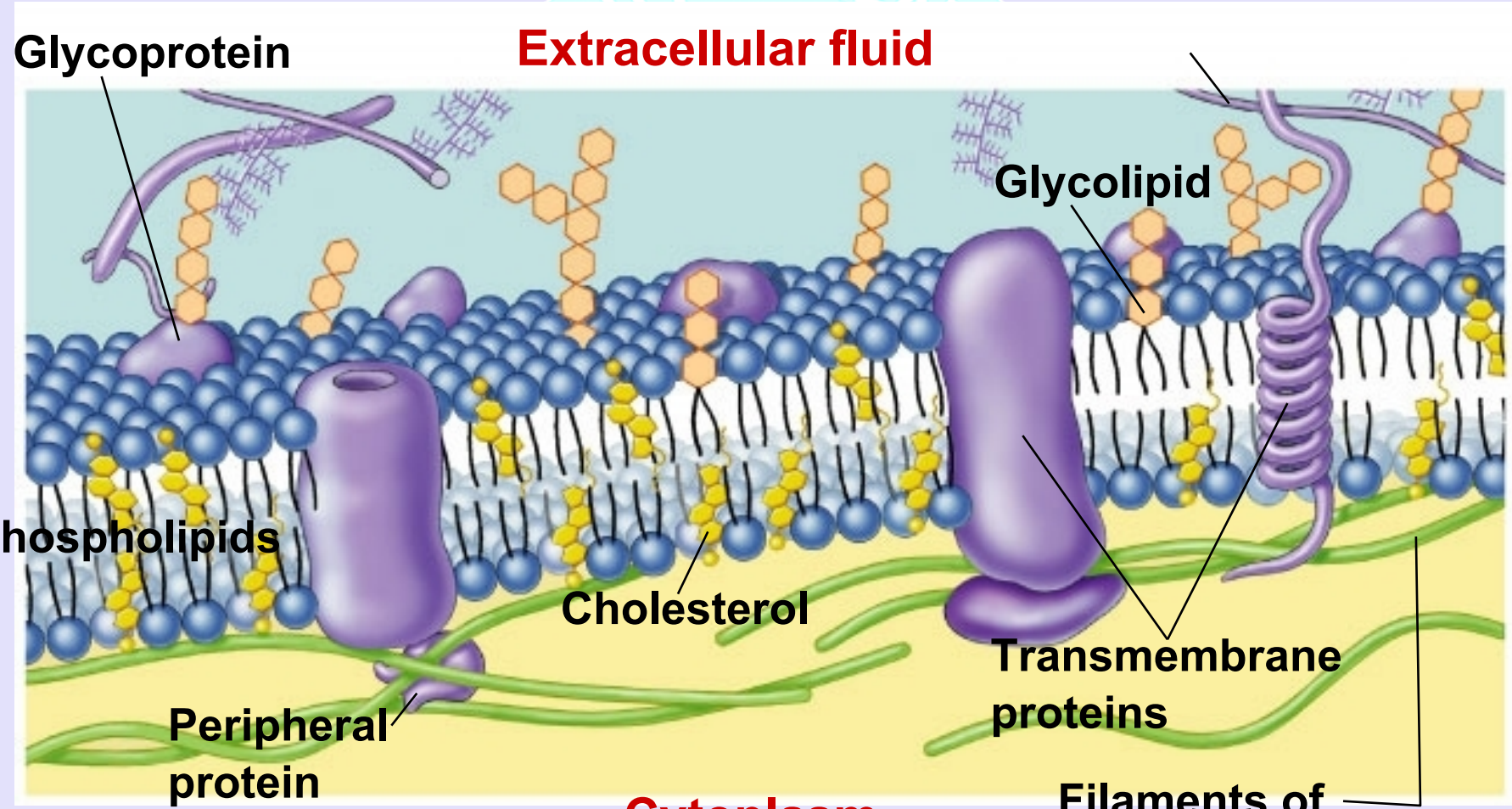
CELL MEMBRANE

(Plasma membrane OR Plasma lemma)

- It surrounds each cell and separating its contents from the extra cellular environment.
- It is selectively permeable and controls the passage of substances into and out of the cell.
- The protoplasm consists of a central more dense part called nucleus, and a peripheral less dense part called the cytoplasm.

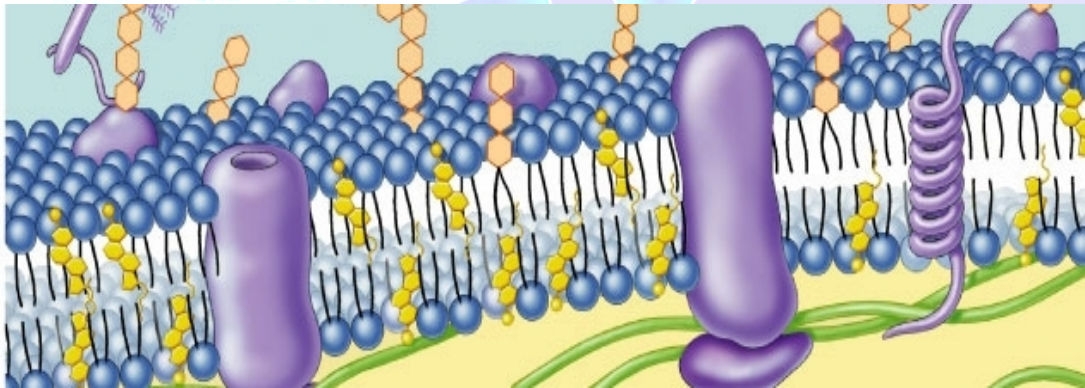


Membrane is a collage of proteins & other molecules embedded in the fluid matrix of the lipid bilayer



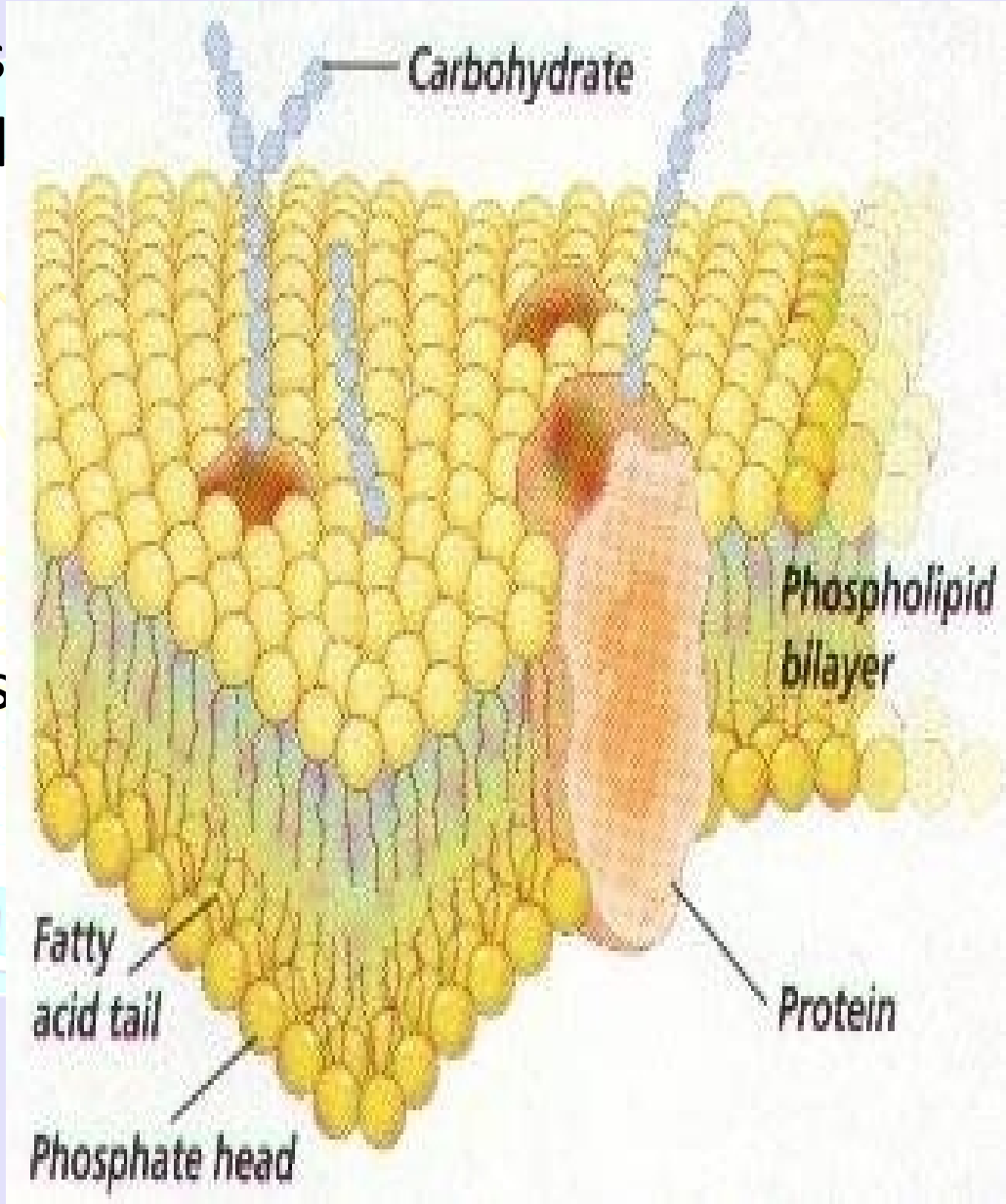
FUNCTIONS OF CELL MEMBRANE

- Cell membrane separates living cell from nonliving surroundings
  - thin barrier = 8 nm thick
- Controls traffic in & out of the cell
  - selectively permeable
  - allows some substances to cross more easily than others
    - hydrophobic v/s hydrophilic
- Made up of phospholipids, proteins & other macromolecules



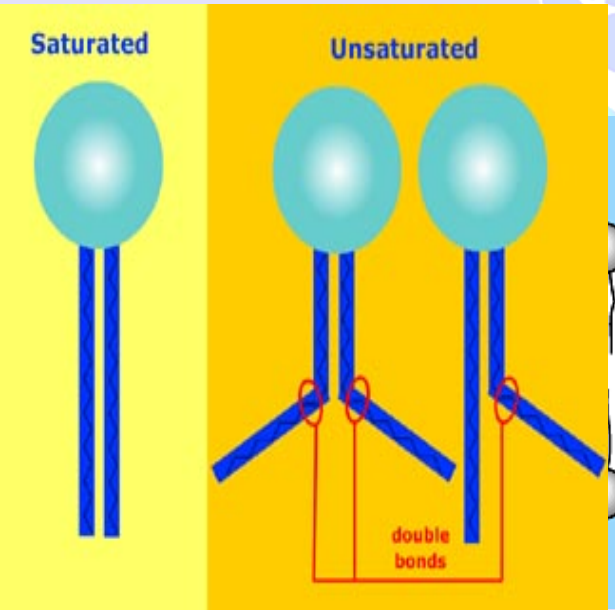
STRUCTURE OF CELL MEMBRANE

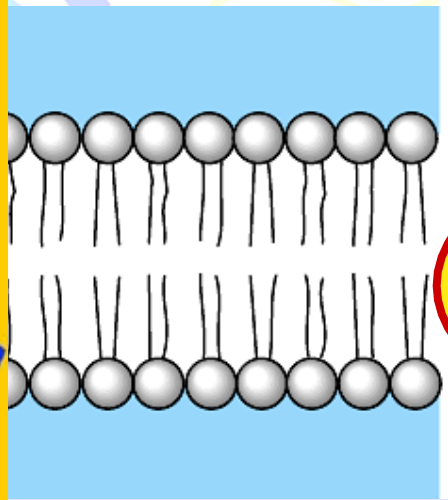
- On biochemical analysis of plasmalemma revealed 35% lipid, including phospholipids and choletrol, 60%proteins and a small amount of carbohydrates
- Cell membrane consists of bilayer of phospholipids

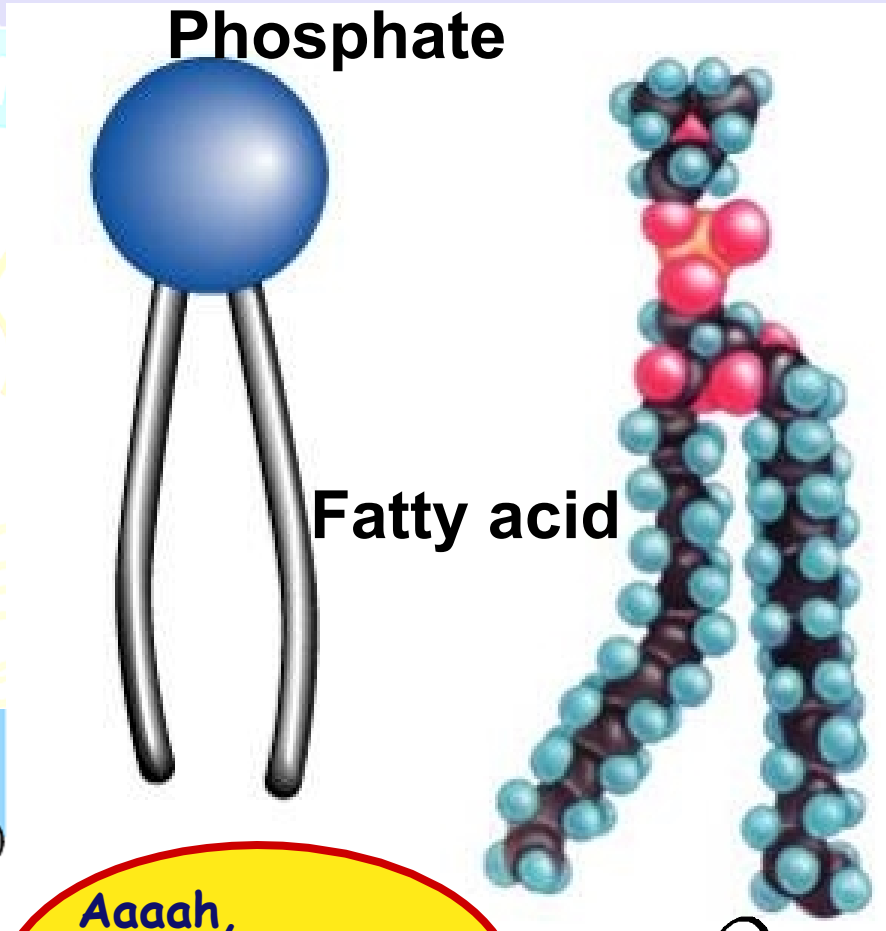


PHOSPHOLIPIDS

- Fatty acid tails
  - hydrophobic
- Phosphate group head
  - hydrophilic
- Arranged as a bilayer







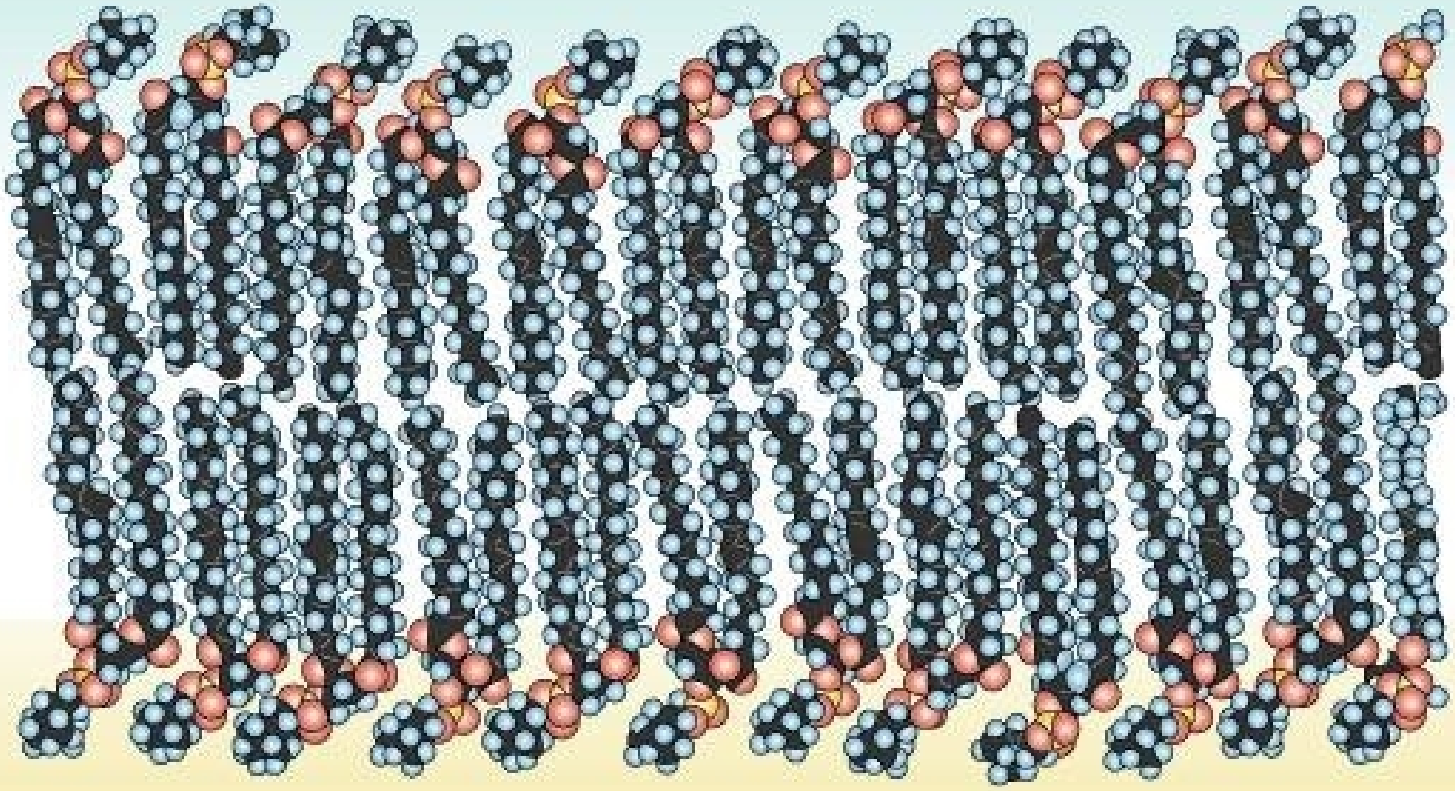


PHOSPHOLIPID BILAYER

polar hydrophilic heads

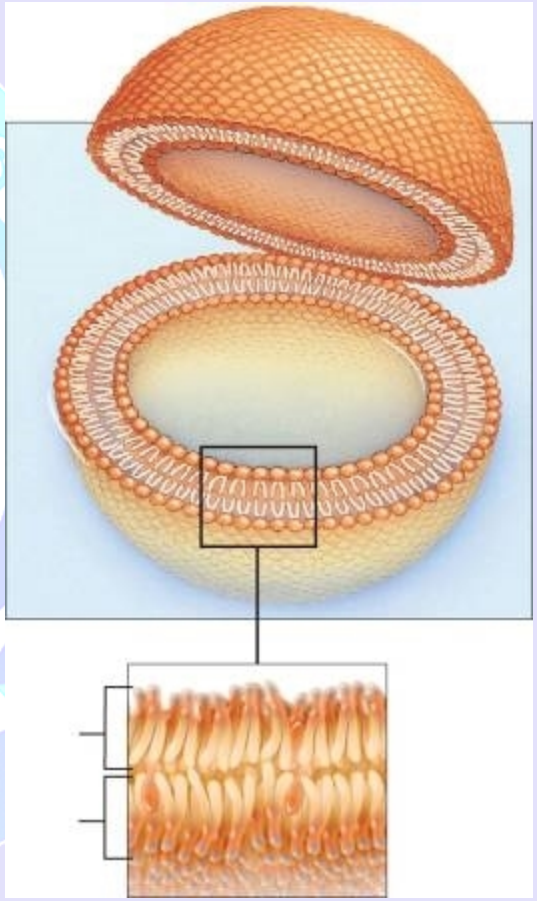
nonpolar hydrophobic tails

polar hydrophilic heads



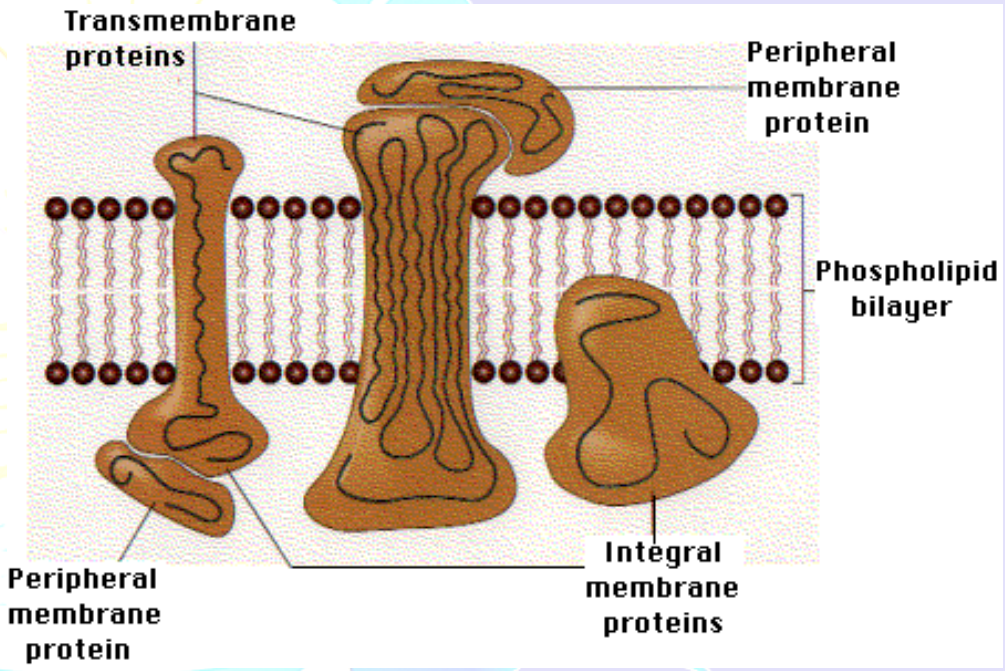
LIPID BILAYER

- Main component of cell membranes
- Gives the membrane its fluid properties
- Two layers of phospholipids



MEMBRANE PROTEINS

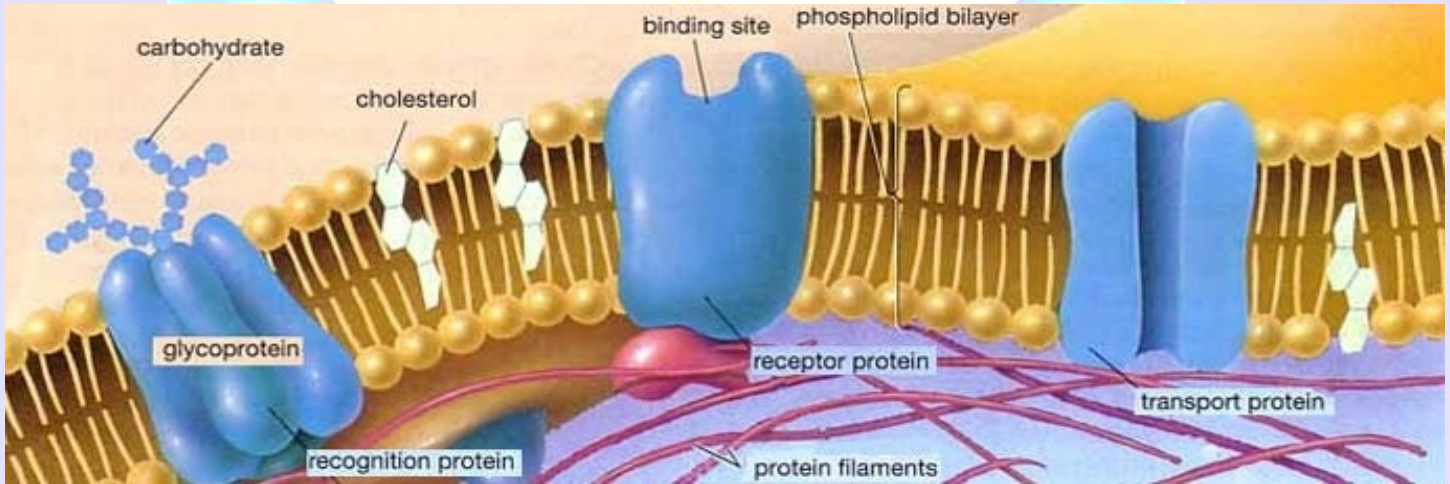
- They are present in the form of irregularly rounded masses.
- Most of them are “Peripheral proteins.”
- Some of them are “Integral or Trans-membrane proteins.”



MEMBRANE PROTEINS

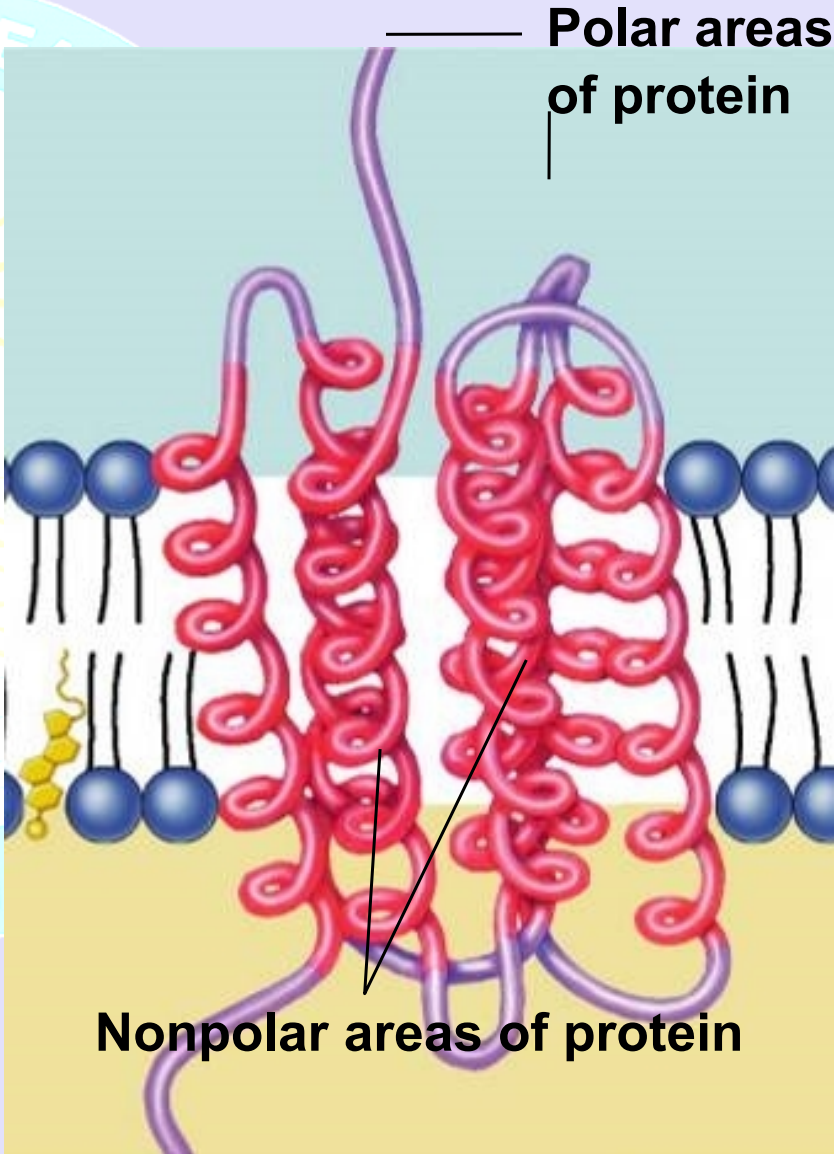
Membrane proteins:

- **Peripheral proteins**
  - **loosely bound to surface of membrane**
  - **cell surface identity marker (antigens)**
- **Integral proteins**
  - **penetrate lipid bilayer, usually across whole membrane**
  - **transmembrane protein**
  - **transport proteins**
    - **channels, permeases (pumps)**



PROTEINS DOMAINS ANCHOR MOLECULE

- Within membrane
  - nonpolar amino acids
    - hydrophobic
    - anchors protein into membrane
- On outer surfaces of membrane
  - polar amino acids
    - hydrophilic
    - extend into extracellular fluid & into cytosol



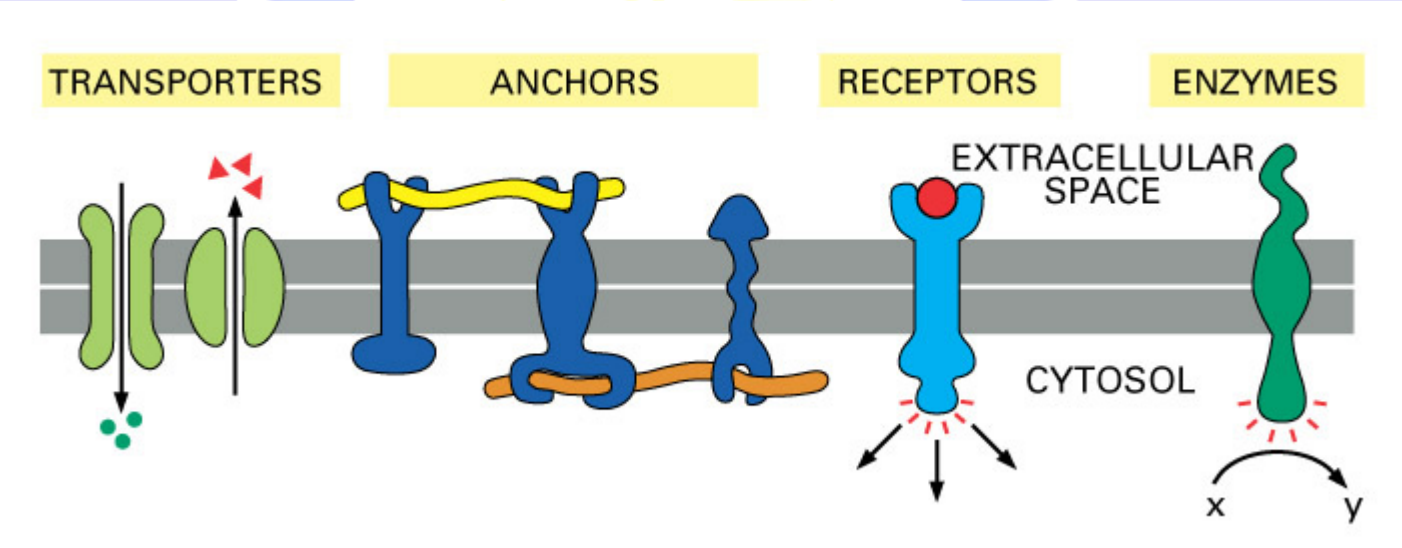
FUNCTIONS OF PROTEIN

- **Significance of proteins:-**
- 1. **Some proteins play vital role in transport across the membrane, and act as pump. Ions get attached to the protein on one surface and move with protein to the other surface.**

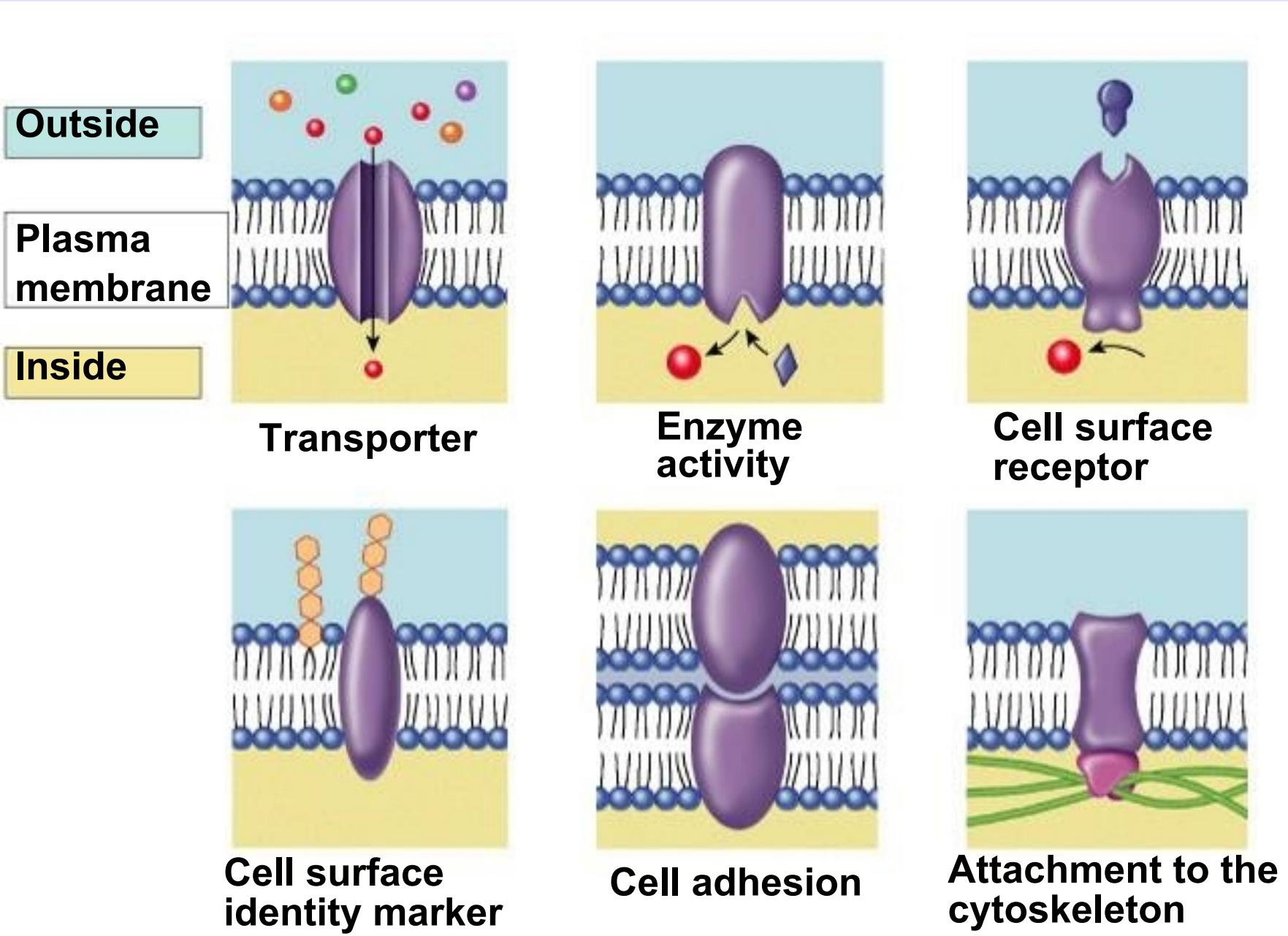
2. Some proteins form passive channels through which substances can diffuse through the membrane.

3. Some proteins act as receptors for specific neurotransmitters or hormones.

4. Some proteins act as enzymes.

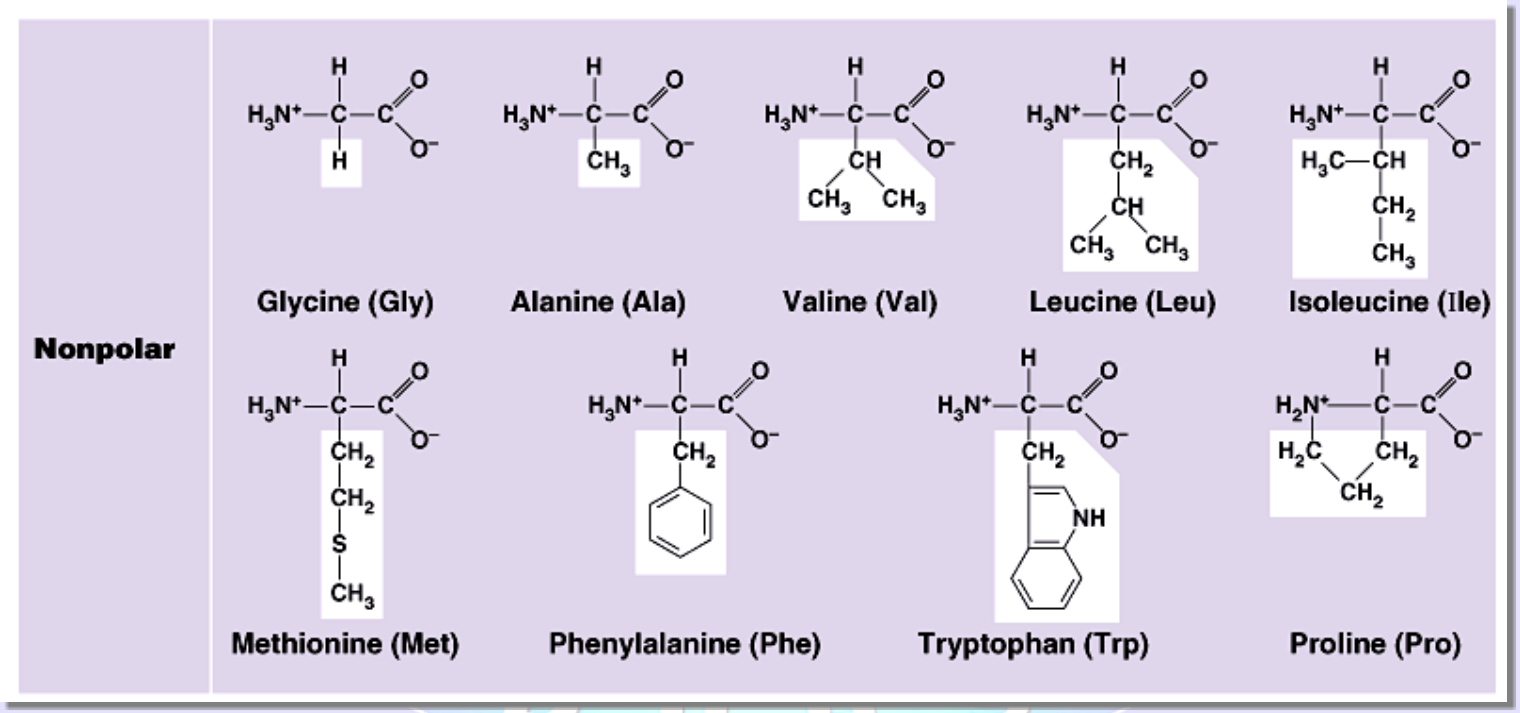


Many Functions of Membrane Proteins



Classes of amino acids

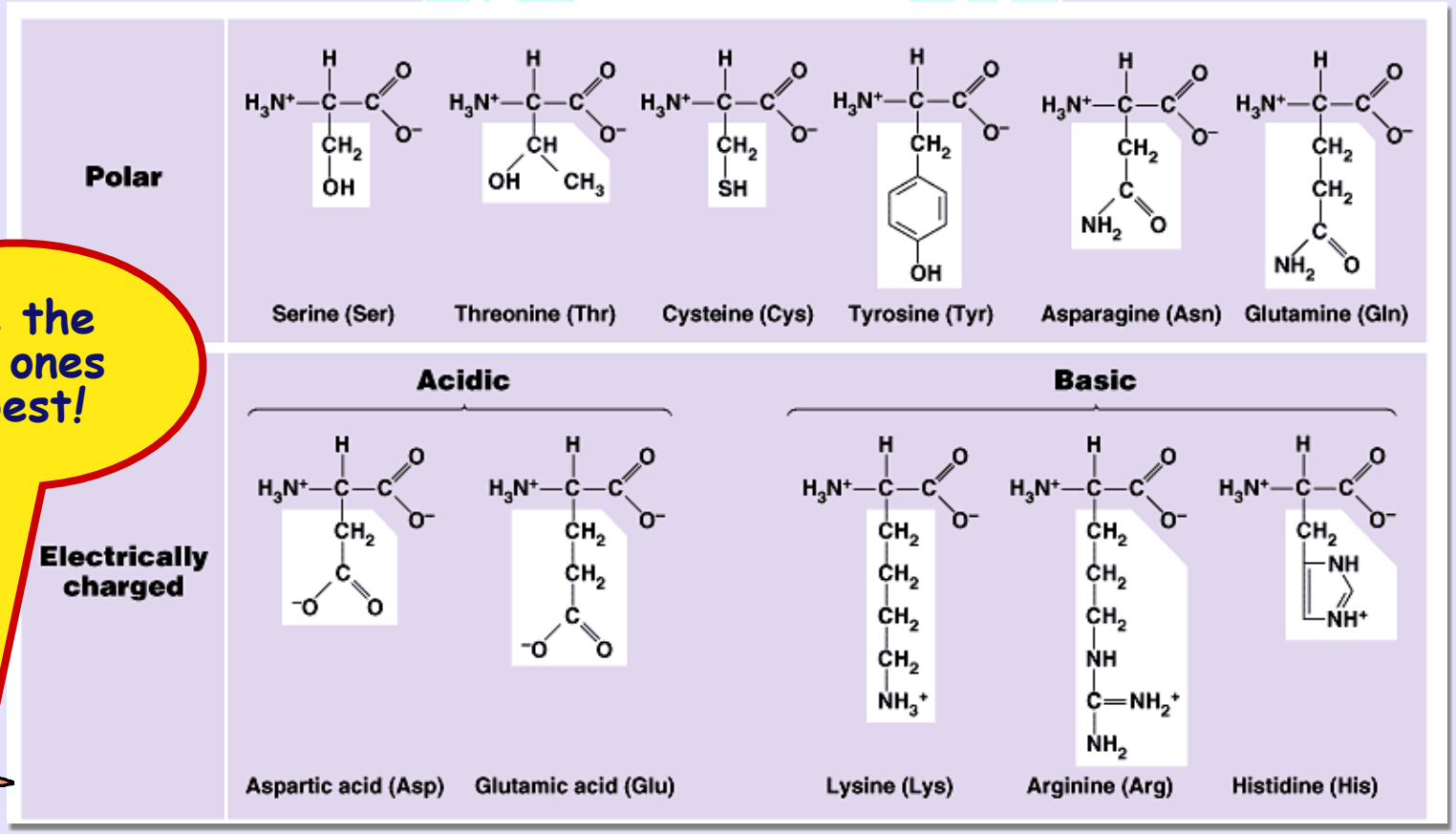
What do these amino acids have in common?

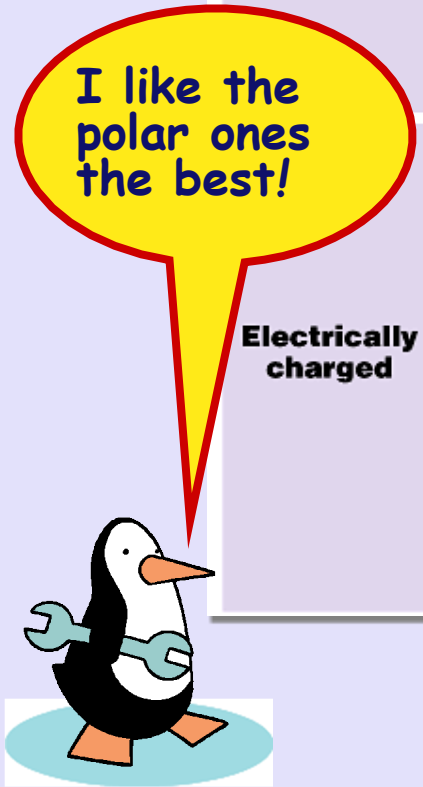


nonpolar & hydrophobic

Classes of amino acids

What do these amino acids have in common?

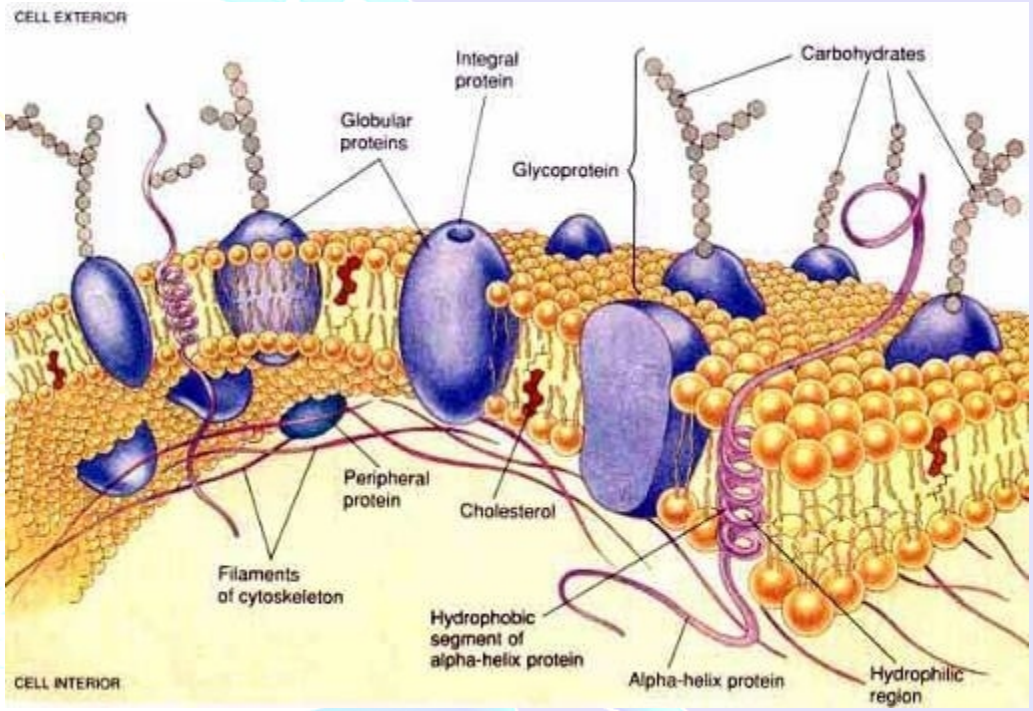




polar & hydrophilic

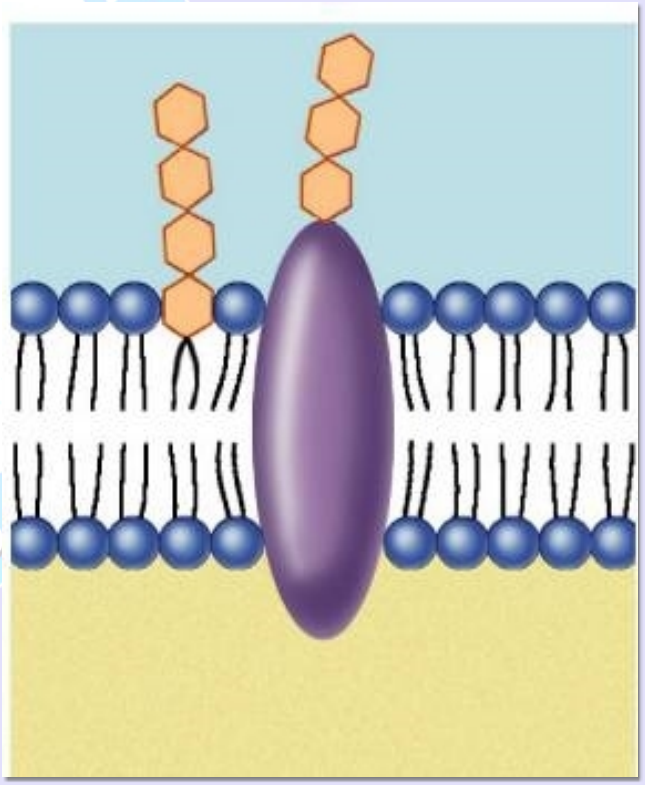
CARBOHYDRATES

- They are present at the surface of cell membrane.
- They are attached either to the proteins (forming glycoproteins), or to the lipids (forming glycolipids).
- Carbohydrates layer is specially well developed on the external surface of membrane, where it called Glycocalyx or Cell coat. Its thickness is vary from 2 to 20 nm.



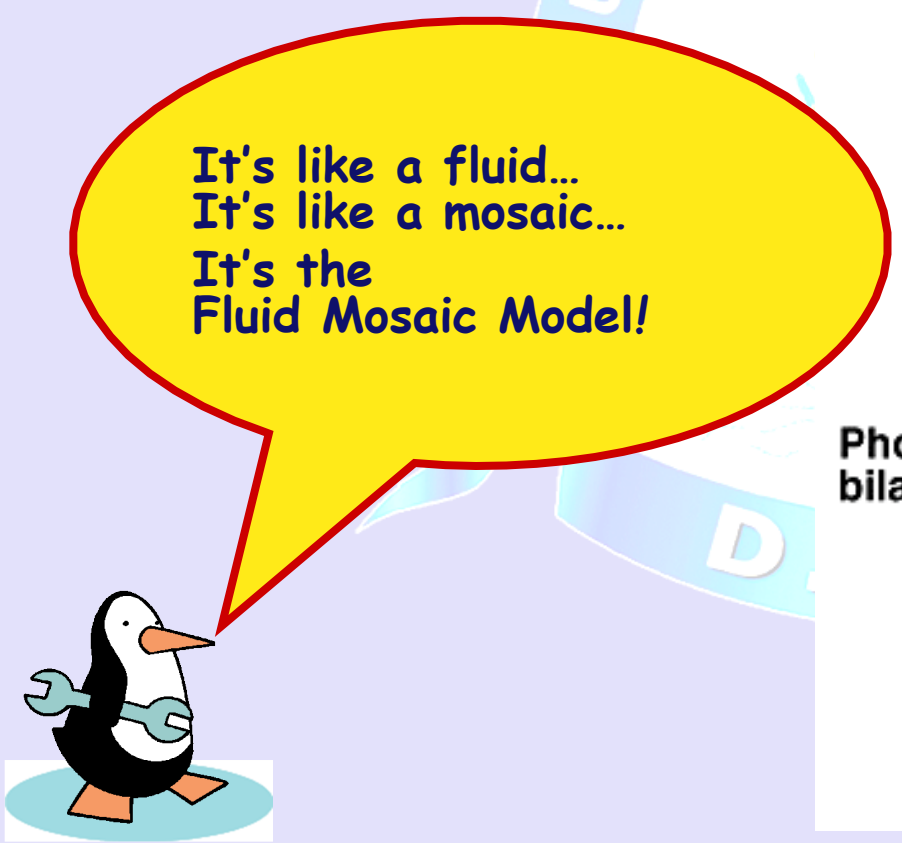
FUNCTIONS OF MEMBRANE CARBOHYDRATES

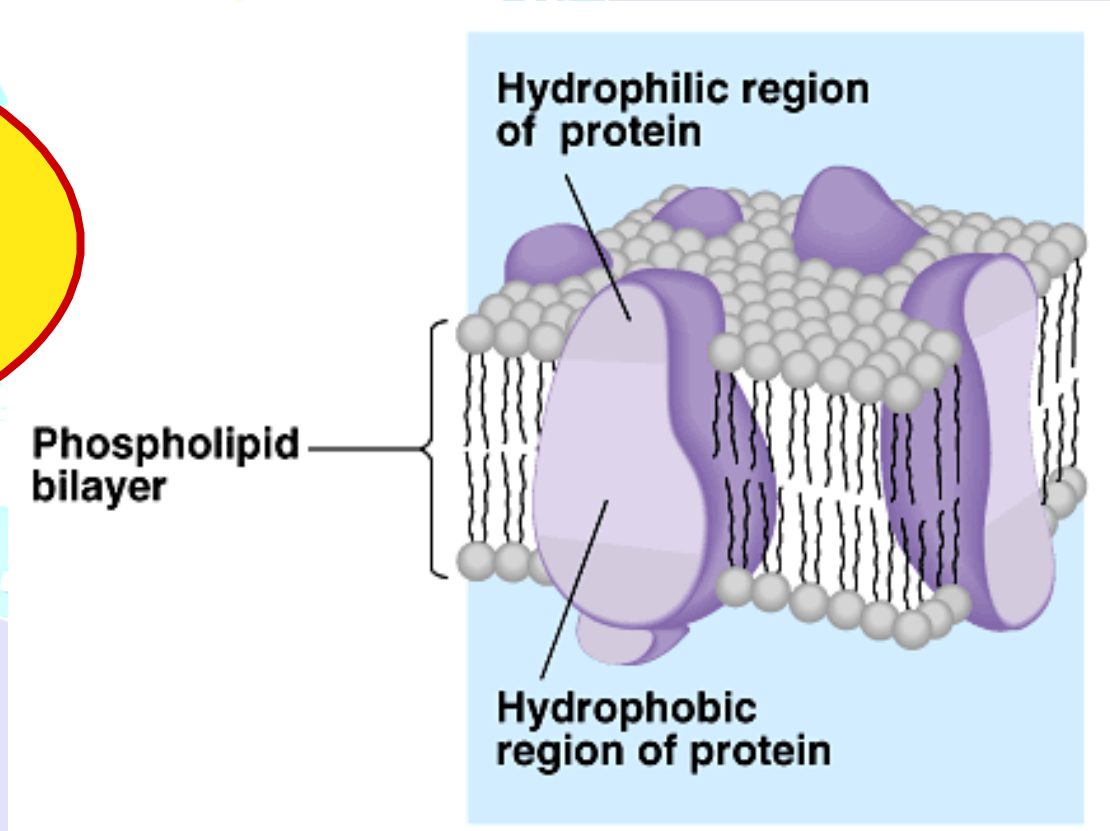
- Play a key role in cell-cell recognition
  - ability of a cell to distinguish one cell from another
    - antigens
  - important in organ & tissue development
  - basis for rejection of foreign cells by immune system



FLUID MOSAIC MODEL

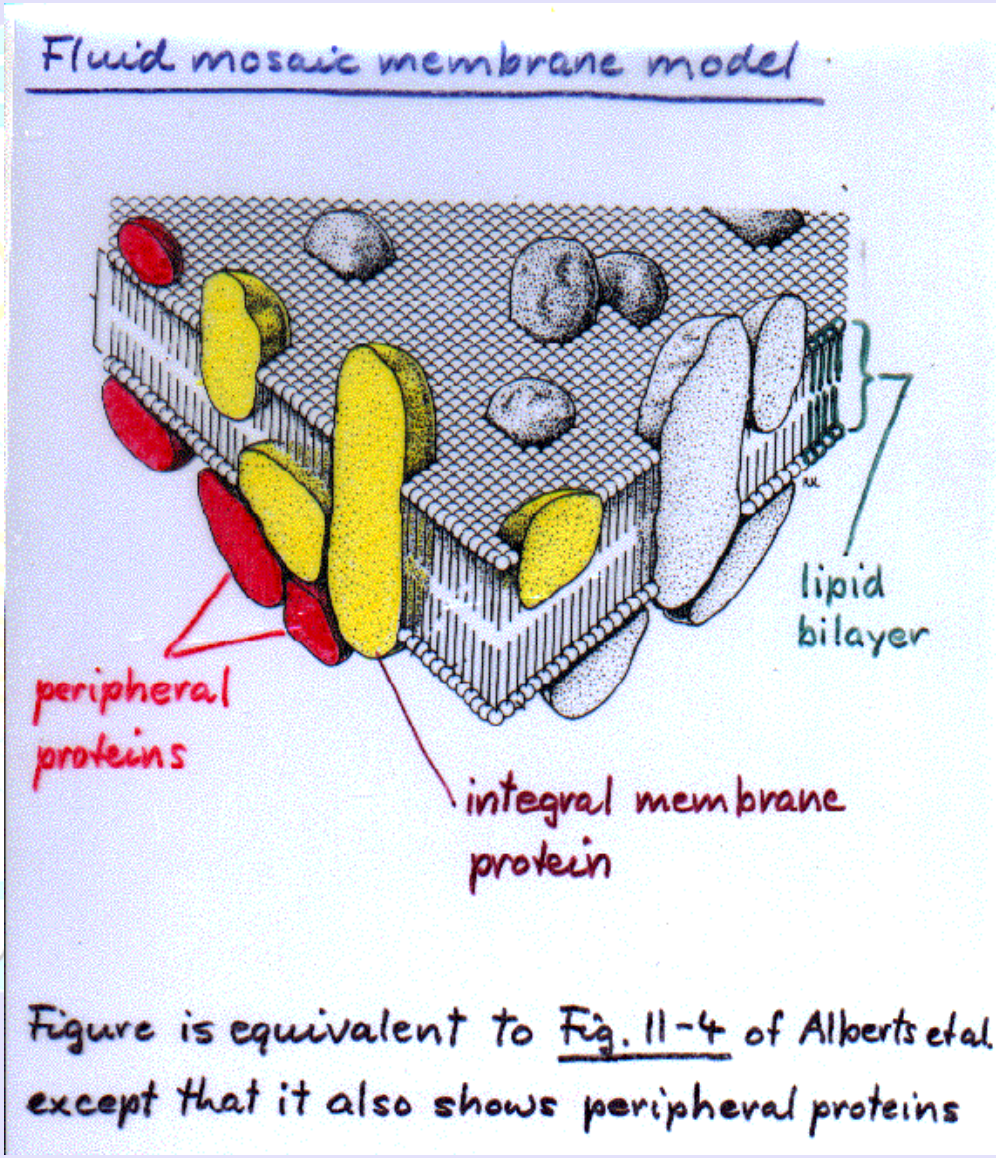
- Membrane is a mosaic of
  - Phospholipids
  - Glycolipids
  - Sterols
  - Proteins
- In 1972, S.J. Singer & G. Nicolson proposed that membrane proteins are inserted into the phospholipid bilayer





STRUCTURE OF CELL MEMBRANE

- On the external surface of plasma membrane of animals cells many of the membrane proteins and some of membrane lipids are conjugated with polysaccharides on the outer surface of cell membrane called as Glycocalyx (cell coat).
- This layer protects the cell membrane.



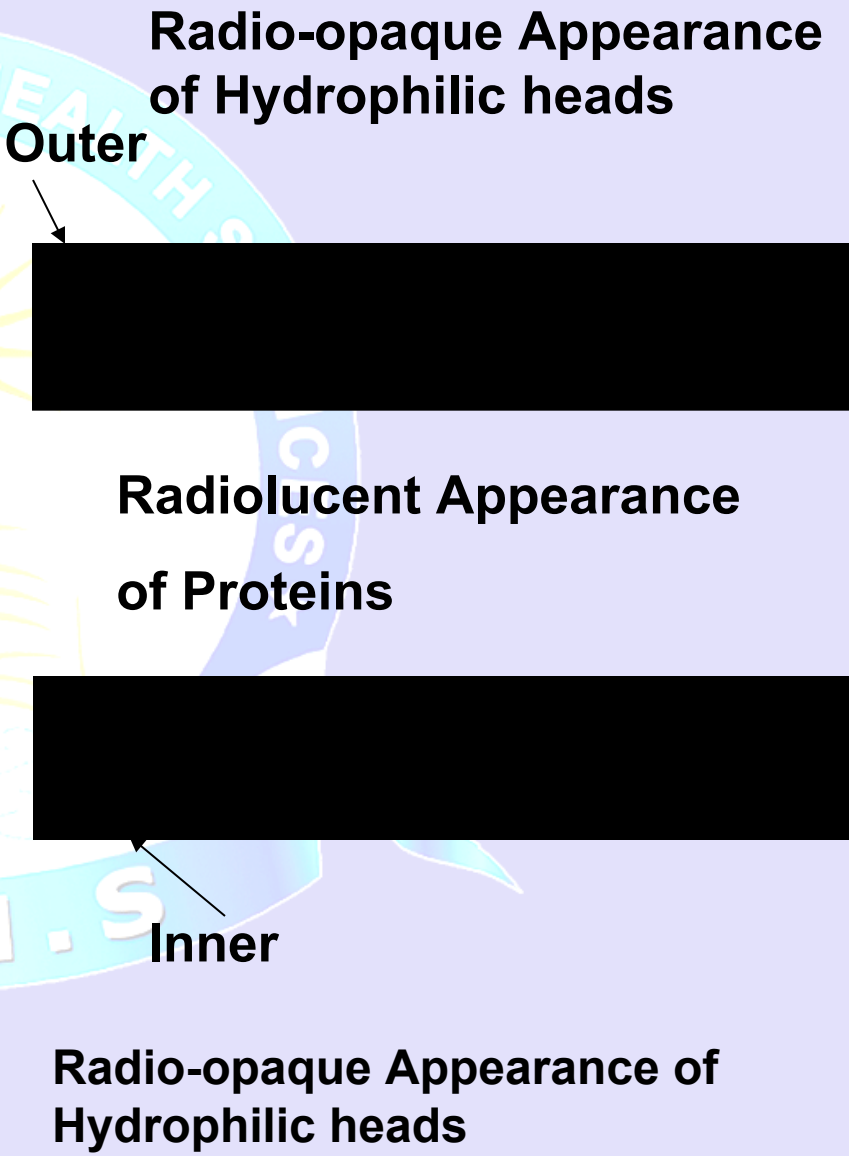
FLUID MOSAIC MODEL

- *Endocytosis* and *exocytosis* handle the really big molecules, such as long protein chains or ringed structures, as well as the bulk volume of small molecules.

- In exocytosis, waste products or hormones, which are contained in vacuoles or vesicles, exit the cell and their containing membrane is absorbed and added to the cell membrane.

Electron Microscopic Appearance of Cell Membrane

- Singer and Nicholson, in early 1970s proposed the Fluid Mosaic Model of membrane structure.
- On electron microscope examination of cell membrane, exhibits a trilaminar structure; there are two electron dense lines and separated by an electron lucent zone
- These arrangement is found not only in cell membrane but also all the organelles present inside the cell. This trilaminar structure is also called as Unit Membrane.



Electron Microscopic Appearance of Cell Membrane

Outer Radio-opaque Layer



Central Radio-lucent Layer



ENDOCYTOSIS

- In endocytosis, substances such as food are brought into the cell in a process in which the cell membrane surrounds the particle and moves the particle inside the cell, creating a vacuole or vesicle as a membrane-enclosed container.

- There are three types of endocytosis:

ENDOCYTOSIS

- ***Pinocytosis*** occurs when the cell absorbs fluid from the exterior, creating a fluid vacuole.
- ***Receptor-mediated endocytosis*** is a special type of pinocytosis that is activated by the identification of a receptor protein sensitive to the specific substance.
- ***Phagocytosis*** is the engulfing and digesting of substances, usually food, by vacuoles with a lysosome attached (a lysosome is an organelle that contains digestive enzymes).

REFERENCES

- Basic histology by Junqueira Page # 24-29.

THANK YOU